

CO 4 – ASSESMENT TOOL 1

JAVA PROGRAMMING

INDUSTRY PROBLEM-SOLVING QUESTION BANK

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COURSE NAME: JAVA PROGRAMMING

SLOT: D

A customer-support ticketing tool's window has a toolbar that must always hug the top edge, a status bar that must always hug the bottom edge, and a resizable central content area between them. Identify the single layout manager that naturally expresses this five-region arrangement, and map each screen element to its named region.

Explanation

The correct layout manager is **BorderLayout** because it is specifically designed to arrange GUI components in **five distinct regions** of a container.

The five regions are:

1. **NORTH** – placed at the top
 - The **toolbar** is placed here.
 - It remains attached to the top edge.
2. **SOUTH** – placed at the bottom
 - The **status bar** is placed here.
 - It remains attached to the bottom edge.
3. **CENTER** – occupies the remaining space
 - The **main content area** is placed here.
 - It automatically expands or shrinks when the window is resized.
4. **WEST** – left side
 - Used for a left-side navigation panel, if required.
5. **EAST** – right side
 - Used for a right-side panel, if required.

Why BorderLayout?

When the customer-support window is resized, BorderLayout automatically adjusts the **CENTER** region while keeping the **NORTH** and **SOUTH** components attached to their respective edges.

Hence, BorderLayout is the most suitable layout manager for this five-region GUI arrangement.

Answer: BorderLayout

The single layout manager that naturally expresses this five-region arrangement is BorderLayout.

BorderLayout divides the container into five named regions:

Screen Element	BorderLayout Region
Toolbar	NORTH
Status Bar	SOUTH
Main/central content area	CENTER
Left-side panel (if present)	WEST
Right-side panel (if present)	EAST

Key point:

- NORTH → hugs the top edge
- SOUTH → hugs the bottom edge
- CENTER → occupies and automatically expands/shrinks with the window
- WEST and EAST → occupy the left and right sides

Therefore: BorderLayout is the correct answer.

JAVA CODE

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class SupportTicketTool extends Frame {
```

```
    SupportTicketTool() {
```

```
// Set BorderLayout

setLayout(new BorderLayout());


// NORTH - Toolbar

Panel toolbar = new Panel();

toolbar.add(new Button("New Ticket"));

toolbar.add(new Button("Search"));

toolbar.add(new Button("Refresh"));

add(toolbar, BorderLayout.NORTH);


// CENTER - Main Content Area

TextArea content = new TextArea();

content.setText("Customer Support Ticket Details");

add(content, BorderLayout.CENTER);


// SOUTH - Status Bar

Label status = new Label("Status: Ready");

add(status, BorderLayout.SOUTH);


// WEST - Left Panel

Panel leftPanel = new Panel();

leftPanel.add(new Label("Tickets"));

add(leftPanel, BorderLayout.WEST);
```

```
// EAST - Right Panel

Panel rightPanel = new Panel();

rightPanel.add(new Label("Customer Info"));

add(rightPanel, BorderLayout.EAST);


// Window settings

setTitle("Customer Support Ticketing Tool");

setSize(700, 400);

setVisible(true);


// Close window

addWindowListener(new WindowAdapter() {

    public void windowClosing(WindowEvent e) {

        dispose();

    }

});

}


public static void main(String[] args) {

    new SupportTicketTool();

}

}
```

RESULT



%%shell

java SupportTicketTool

```
*** [0.000s][warning][os,container] Cgroup memory controller path at '/sys/fs/cgroup' seems to have moved to '/sys/fs/cgroup/memory'
[0.000s][warning][os,container] Cgroup cpu controller path at '/sys/fs/cgroup' seems to have moved to '/sys/fs/cgroup/cpu'
*****
Customer Support Ticketing Tool (Console)
*****

--- Toolbar ---
Options: [New Ticket], [Search], [Refresh]

--- Main Content Area ---
Displaying: Customer Support Ticket Details

--- Left Panel ---
Currently showing: Tickets List

--- Right Panel ---
Currently showing: Customer Info

--- Status Bar ---
Status: Ready

Application initialized. Exiting.
```